INSTRUCTIONS

GEK-36457C

VIBRATION DETECTORS

CATALOG NO. 8560K12

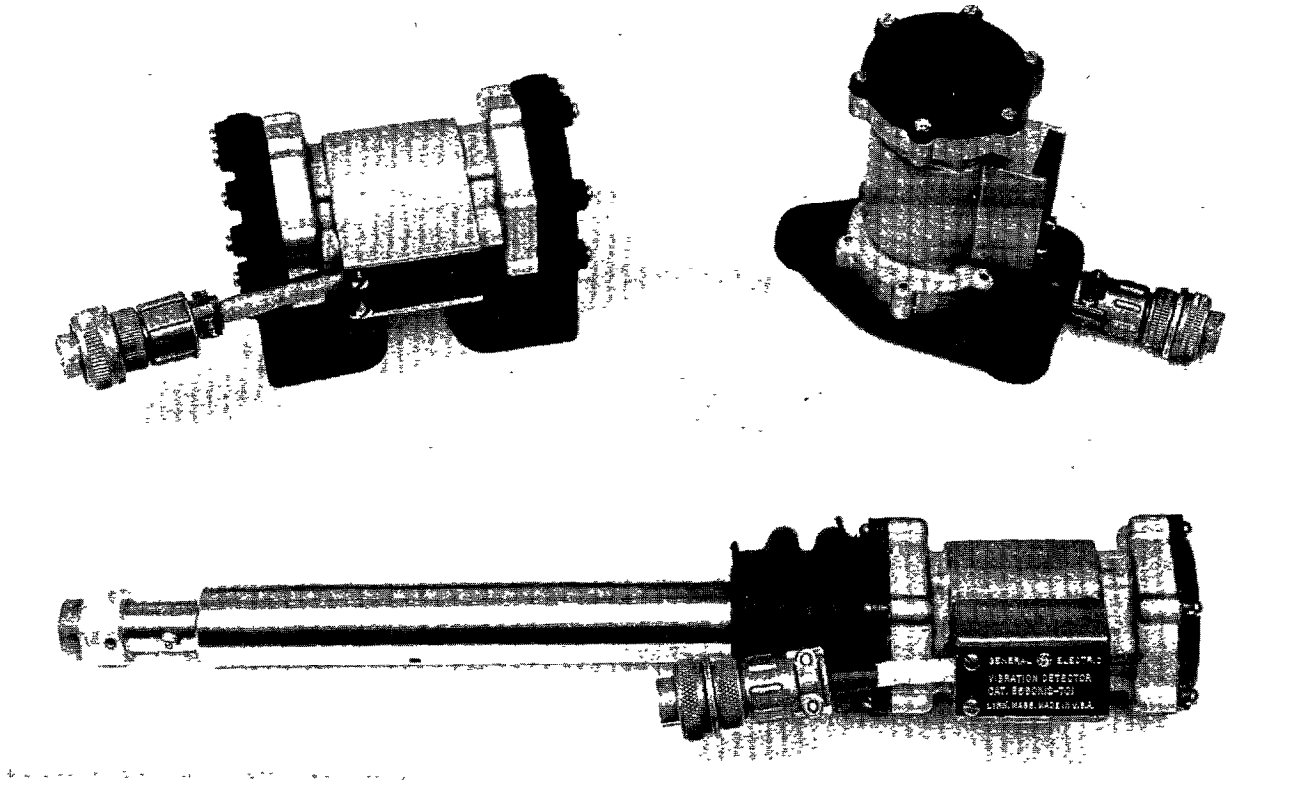


Figure 1 - Vibration Detectors (Cat. No. 8560K12)

INTRODUCTION

The General Electric 8560K12 vibration detectors (shown in Figure 1) are velocity-type detectors with an output of 0.4 volts rms per inch per second of rms vibration velocity. The frequency response is essentially flat from 13 hz. up; that is, the ratio of the output voltage to the frequency of vibration (for a given amplitude of vibration) is practically constant throughout the frequency range.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

DIFFERENCE DATA

GEK-36514
(Supplement to GEK-36457)

VIBRATION DETECTORS

Catalog Number 8560K12G800 through -808

THIS SHEET MUST BE USED WITH PUBLICATION GEK-36457

Vibration detectors cat. no. 8560K12G800 through -808 are similar to the detectors described in publication GEK-36457, except for the differences listed in the supplement to Table 1.

The 750F detectors marked with an asterisk (*) in the supplement to Table 1 are no longer manufactured. For recommendations for replacement detectors for these types, refer to the General Electric Installation and Service Engineering Office responsible for the gas turbine on which the detector is used.

SUPPLEMENT TO TABLE 1

Vibration Detectors, Cat. No. 8560K12

Detector Identification	Maximum Recommended Temperature, °F, (Detector)	Type Mounting	Sensitive Axis, Relative to Mounting Surface	Three-Inch Stub-Cable & Plug Included	Remarks
8560K12G800	400	Stud	Perpendicular	No	With 15 Ft. 400F shielded cable and special nameplate.
-801*	750	Stud	Perpendicular	No	With 30 Ft. 750F shielded cable and special nameplate
-802*	750	Stud	Perpendicular	No	With 60 Ft. 750F shielded cable and special nameplate
-803*	750	Stud	Perpendicular	No	With 40 Ft. 750F shielded cable and special nameplate
-804*	750	Stud	Perpendicular	No	With 70 Ft. 750F shielded cable and special nameplate
-805	400	Stud	Perpendicular	No	With 60 Ft. 400F shielded cable and special nameplate
-806	400	Stud	Perpendicular	No	With 30 Ft. 750F shielded cable and special nameplate
-807	400	Stud	Perpendicular	No	With 70 Ft. 750F shielded cable and special nameplate
-808	400	Stud	Perpendicular	No	With special nameplate. No cable included

*No longer available

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GENERAL ELECTRIC

6/76 (400)

INTRODUCTION (continued)

At frequencies below 13 hz, the response falls off rapidly, dropping to approximately one-half output at 9 hz.

The vibration detectors are sealed to exclude dust and oil, and are suitable for use at temperatures up to 400F (200C).

The 8560K12 vibration detectors are not recommended for use at vibration frequencies over 150 hz.

The detectors are ordinarily used in conjunction with the following General Electric equipments:

- 5482401-449 turbine supervisory instrument systems
- 4980175 vibration-amplitude recording equipment
- 5470387 and 5797625 vibration relays
- 5797653 multichannel vibration trip relay
- 1257K18 multichannel vibration amplifier
- 50-390010AAAA1 - 50-390020ZZZZ1 turbine supervisory instrument systems
- 3S390010AAAA1 - 3S390020ZZZZ1 turbine supervisory instrument systems

The detectors may, however, be used in other applications requiring a detector of this type, for example,in conjunction with oscilloscopes, balancing equipment, etc.

For complete operating instructions for the Cat. No. 5482401 - 449 turbine supervisory instrument systems, refer to the applicable system instructions. For the 4980175 vibration-amplitude recording equipment, refer to the instructions GEI-36807 or GEI-73417. The 5470387 vibration relay is described in instructions GEI-56464, the 5797625 vibration relay in instructions 4542K13-001. For description of the 5797653 multichannel vibration trip relay, refer to instructions 4542K12-001. Refer to 4542K14-001 instructions for description of the 1257K18 multichannel vibration amplifier. For the 50-390010AAAA1 - 50-390020ZZZZ1 and 3S390010AAAA1 - 3S390020ZZZZ1 turbine supervisory instrument systems, refer to the title and contents page of the applicable turbine supervisory instrument systems instructions.

RECEIVING, HANDLING AND STORAGE

On receipt of the vibration detectors, examine them immediately for any damage that might have occurred in transit. Immediate examination is especially important if rough handling is evident. If damage is found, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified immediately.

Handle the detectors carefully when unpacking and installing them to prevent damage to the coil and suspension springs inside the detectors. DO NOT DROP.

RECEIVING, HANDLING AND STORAGE (continued)

If the detectors are to be stored prior to installation, repack them in their original containers (cushioned in the original packing material) and store in an area free of corrosive fumes, high humidity, and temperature extremes.

DESCRIPTION

FUNCTIONAL DESCRIPTION

A cross-sectional view of the vibration detector is shown in Figure 2. A coil, wound on an aluminum cup, is seismically suspended in the field of a permanent magnet by means of two flat three-point suspension springs. When the case and permanent-magnet assembly is vibrated, the coil, being seismically suspended, does not follow the vibratory motion, but rather stays motionless in space. Hence, an alternating voltage is generated by the alternating motion of the magnetic flux past the coil. The suspended coil is damped by eddy currents generated in the aluminum cup upon which the coil is wound. The double-headed arrow on the case indicates the component of vibration being measured.

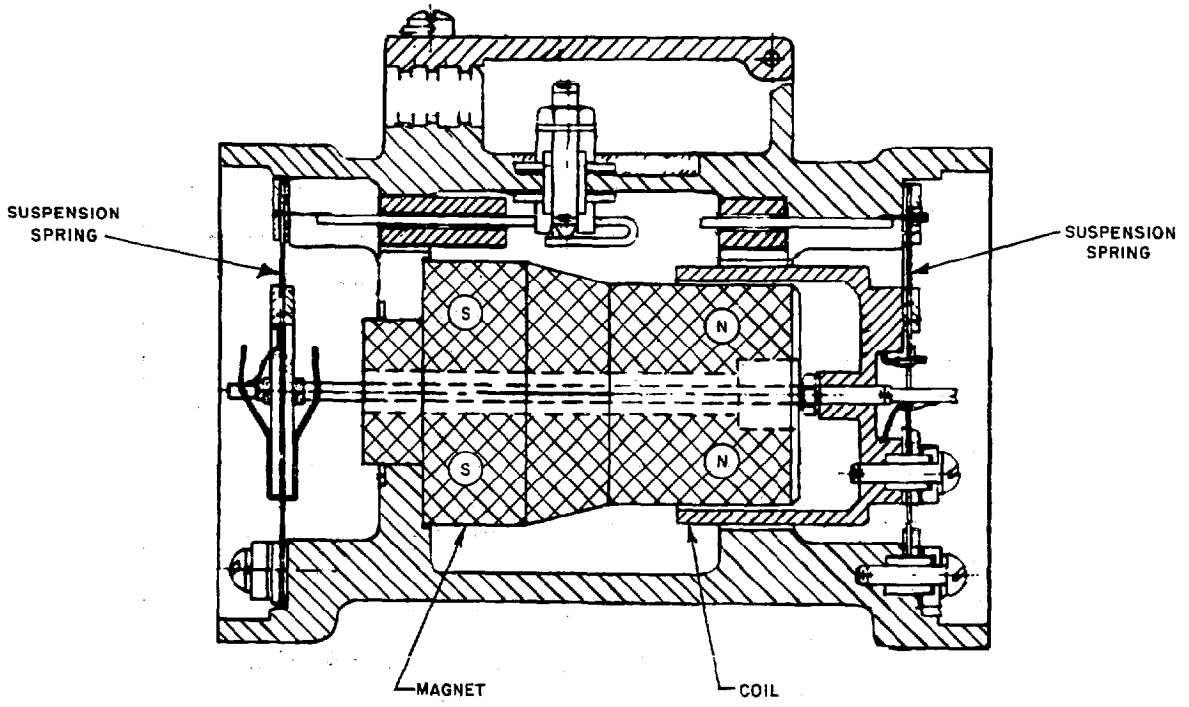


Figure 2 - Cross-sectional view of vibration detector (end covers not shown)

DESCRIPTION OF TYPES

The various types of vibration detectors are listed in Table I. Detectors 8560K12G702 thru G707 are furnished with a three-inch cable and plug providing a quick means of connecting and disconnecting. Other cables, if required, are ordered as a separate item (see Table II). Probes, if required, are also ordered as a separate item (see Table III).

DESCRIPTION (continued)

TABLE I - VIBRATION DETECTOR (8560K12)

Detector Identification	Maximum Recommended Temperature °F (Detector)	Type Mounting	Senstive Axis (Relative to Mounting Surface)	Three-Inch Stub-Cable & Plug Included	Stub-Cable Temperature Rating °F	Remarks
8560K12G700	750	stud	perpendicular	no	- -	No longer available
701	400	stud	perpendicular	no	- -	
702	400	stud	perpendicular	yes	250	For stud-mounting or head for probe-type detector. Rubber boot for probe is included.
703	400	bracket	perpendicular	yes	250	
704	400	bracket	parallel	yes	250	
705	400	stud	perpendicular	yes	400	
706	400	bracket	perpendicular	yes	400	
707	400	bracket	parallel	yes	400	

NOTE: For detectors cat. no. 8560K12G800 and up, see supplementary instructions, GEK-36514.

TABLE II - CABLES

THREE-INCH STUB CABLE

250F -- 44A370935-G01

400F -- 44A370935-G02

Furnished with eyelet terminals at detector end and MS-3101A-12S-3P plug on other end.

EXTENSION CABLES

5794543	Max. Temperature (cable) °F	Length Feet
G1	250	5
G2	250	10
G3	250	15
G4	250	20
G5	250	25
G6	250	50
G7	400	5
G8	400	25
G9	400	15
G10	400	40
G11	250	30
G12	250	40
G13	250	60

Above cables are furnished complete with receptacle MS-3106A-12A-3S (mates with MS-3101A-12S-3P).

750F CABLE

44A370933	Length Feet
-G01	2
-G02	5
-G03	10
-G04	15
-G05	25
-G06	30
-G07	50
-G08	60
-G09	40
-G10	70

Furnished with eyelet terminals for detector end (no plug)

400F CABLE

44A370934	Length Feet
-G01	2
-G02	5
-G03	10
-G04	15
-G05	25
-G06	30
-G07	50
-G08	60

Furnished with eyelet terminals for detector end (no plug).

DESCRIPTION (continued)

SHAFT-VIBRATION DETECTORS (See Figure 5)

Detector 8560K12G702 can be equipped with a probe for measuring machine shaft vibration. The probe contains a spring-loaded push rod and a rider tip which bears against the rotating shaft, the spring force being approximately 13 pounds. Shaft vibration detectors will follow all the vibratory motion of the shaft as long as the peak of acceleration does not exceed 2.7g (one g = 386 inches/sec.^{2}), which is equivalent to a vibration displacement amplitude of 0.015 inch peak-to-peak at a frequency of 60 hz, in the case of sinusoidal vibration.

Shaft vibration detectors are recommended only for applications where the following values will not be exceeded:

- Maximum vibration frequency = 150 hz
- Maximum longitudinal vibration displacement amplitude = 0.05 inch peak-to-peak
- Maximum longitudinal peak acceleration = 2.7g
- Maximum transverse peak acceleration = 5g

The machine must be specifically built to accommodate a shaft vibration detector, since it is necessary to provide an entrance hole through the bearing of the machine, together with suitable means for holding the probe in the bearing.

The various lengths of probes are listed in Table III.

Some General Electric steam turbines are equipped with aluminum probes, Dwg. 917C348, manufactured by the GE Large Steam Turbine Department. Instructions for the 917C348 probe will be found in Large Steam Turbine Department instruction GEI-88695.

TABLE III - PROBES

Probe Identification	Length Inches (A in Figure 5)	Hole Location, Inches (B in Figure 5)
C5470903G1	10	6½
2	11	7
3	12	6½
4	13	7
5	14	7
6	15	7
7	16	7
8	17	7
9	18	7
10	19	7
11	20	7
12	22	7
13	21	7
14	23	7
15	27	7
16	24	7
17	8	4
18	25	7
19	28	7
20	9	5
21	26	7
22	7	3
23	19	13
24	24	16
25	35	7

DESCRIPTION (continued)

STUD-MOUNTED AND BRACKET MOUNTED DETECTORS (Figures 3,4,6 and 7)

Stud-mounted and bracket-mounted detectors are designed for installation on the bearing cap or other parts of the machine where vibration is to be measured, and include all types listed in Table I, except when equipped with a probe.

Bracket-mounted vibration detectors are recommended only for applications where the following values will not be exceeded:

- Maximum vibration frequency = 150 hz
- Maximum longitudinal vibration displacement amplitude = 0.05 inch peak-to-peak
- Maximum longitudinal peak acceleration = 20g
- Maximum transverse peak acceleration = 5g

INSTALLATION

Handle the detectors carefully to prevent damaging the internally suspended mechanism. DO NOT DROP. Keep detectors packed in their original shipping cartons (cushioned in the original packing material) for protection until ready to begin installation.

SHAFT-VIBRATION DETECTORS

For accurate measurement of shaft vibration, it is essential that the probe be securely held in the bearing of the machine, as extraneous vibration (such as chattering of the probe assembly in its mounts) must be avoided. Holding arrangements for the probe are ordinarily supplied by the manufacturer of the machine on which the detector is to be mounted.

It is recommended that the outer casing of the probe be held at two points, the tip end (as near to the end as practicable and not more than $1\frac{1}{2}$ inches from this end) and the detector end (not more than $1\frac{1}{2}$ inch from the hex nut). For example, the probe casing is often held at the inner end (near the machine shaft) by a close fitting bushing not over 1.000 inch inside diameter, and near the detector end by a split sleeve and compression nut assembly which clamps the tubular casing of the probe and holds it when the compression nut is tightened. The oil bleed-off hole (see Figure 5) should be located inside the machine such that any oil dripping from the holes will drain back into the oiling system.

The detector may be mounted in any position provided the end of the case marked "X" is level with or higher than the unmarked end. This is necessary to assure proper positioning of the suspended coil in its magnet gap. The rider-tip must bear against a portion of the shaft that is continuously lubricated. Under no condition should a shaft rider be operated without adequate lubrication.

To install the detector, insert the probe into the holders on the bearing of the machine. Press the detector in until the rider-tip is felt to contact the shaft, then depress the probe casing until the rider shaft engages the spring. The

INSTALLATION (continued)

SHAFT-VIBRATION DETECTORS (continued)

tip replacement warning mark (see Figure 5) should now be flush with the outer surface of the hex nut of the probe casing, and the installation mark should not yet be visible. Further depress the probe casing into the machine until the installation mark (second mark) is just flush with the outer surface of the hex nut. Tighten the probe securely in the machine.

These detectors are designed so that, when installed as directed above, the rider-tip may safely wear 1/8 inch before it is necessary to replace the tip. When worn 1/8 inch, the rider shaft comes against a stop, preventing further wear and assuring that the metal tip-holder will not contact the shaft. However, when the tip is fully worn, the detector no longer follows the vibration faithfully, since the rider shaft is against the stop. This condition is indicated when the tip replacement warning mark becomes flush with the outer surface of the hex nut. Tighten the probe securely in the machine.

CAUTION

UNDER NO CIRCUMSTANCES SHOULD TIP WEAR BE COMPENSATED FOR BY DEPRESSING THE PROBE CASING FURTHER INTO THE BEARING IN AN ATTEMPT TO AGAIN BRING THE INSTALLATION MARK INTO VIEW. Doing so may score or damage the shaft of the machine upon further wearing of the tip. For similiar reasons, DO NOT INSTALL DETECTORS HAVING A WORN TIP. If it is necessary to re-install a detector, always replace the tip first, unless the old tip is known to be only negligibly worn. Only with a new factory-length rider tip can the probe be safely installed as directed above with safety to the machine shaft.

For instructions on the installation of Dwg. 917C348 probes, manufactured by General Electric Large Steam Turbine Department, refer to GEI-88695.

STUD-MOUNTED AND BRACKET-MOUNTED DETECTORS

Dimensions for the stud-mounted detectors are shown in Figures 3 and 4. The perpendicular bracket-mounted detector is shown in Figure 6, while the parallel bracket-mounted type is shown in Figure 7. Position the detector on the machine so that it will measure vibration in the desired direction. These detectors may be mounted in any position as long as the end of the case marked "X" is level with or higher than the unmarked end. In cases where it is not possible to comply with this, it will be satisfactory (although less desirable) to mount with the "X" end lower than the unmarked end. The direction of sensitivity of the detector is indicated by a double-ended arrow on the transducer case.

For the bracket-mounted detectors, drill and tap 1/4 inch - 20 holes in the surface on which the detector is to be mounted. (See Figures 6 and 7.) The mounting surface must be flat. Fasten the detector to the machine with the cap screws and lockwashers furnished.

INSTALLATION (continued)

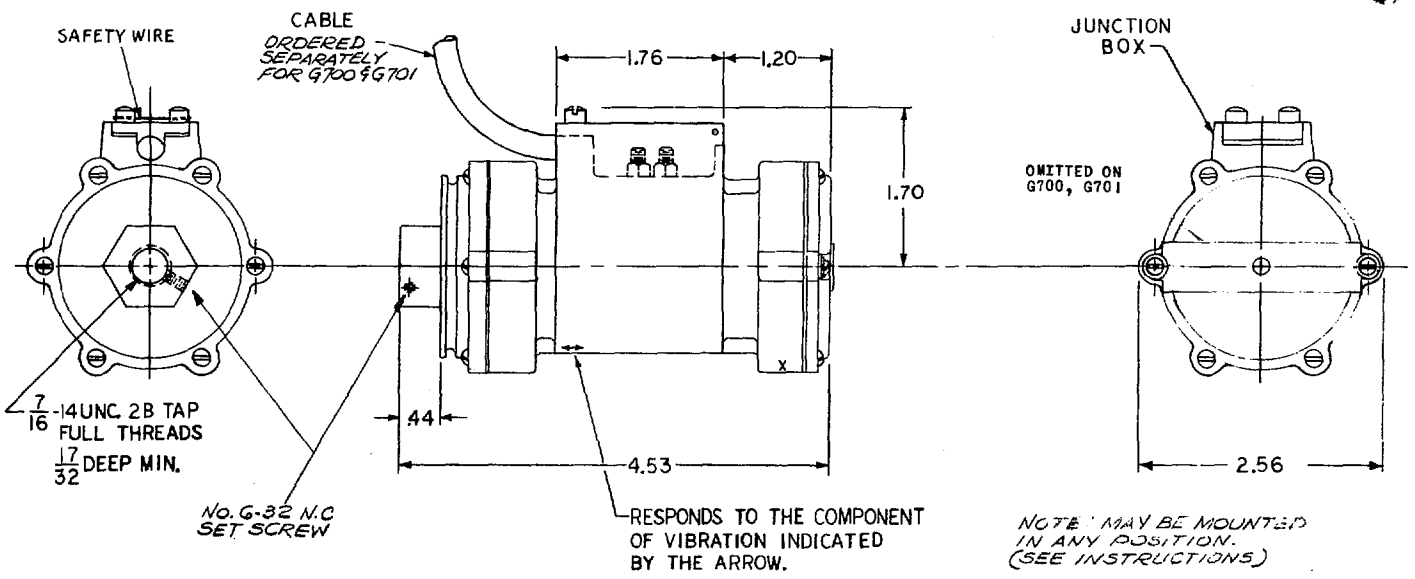


Figure 3 - Outline Dimensions for the 8560K12G700, 701, 800, 801, 802, 803 and 804 Vibration Detectors

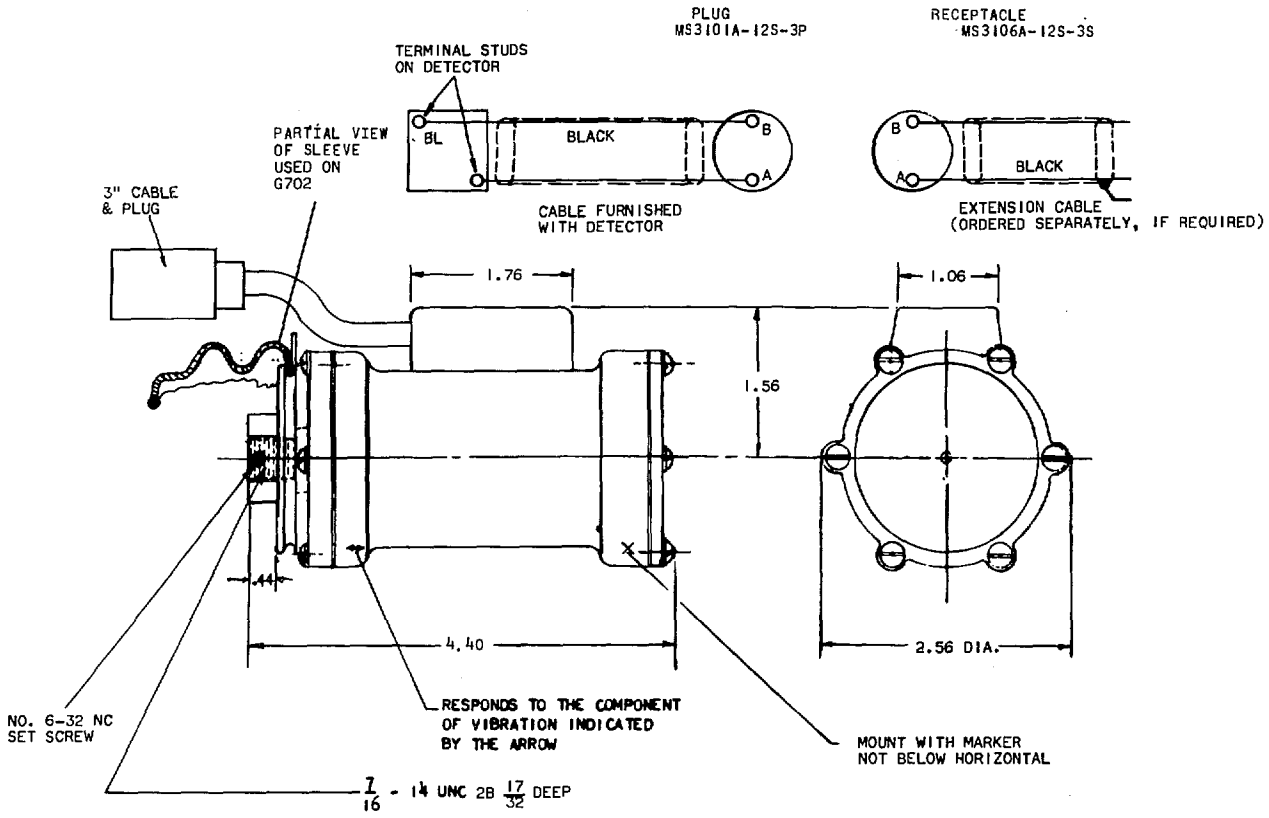


Figure 4 - Outline Dimensions for the 8560K12G702 and 705 Vibration Detectors

INSTALLATION (continued)

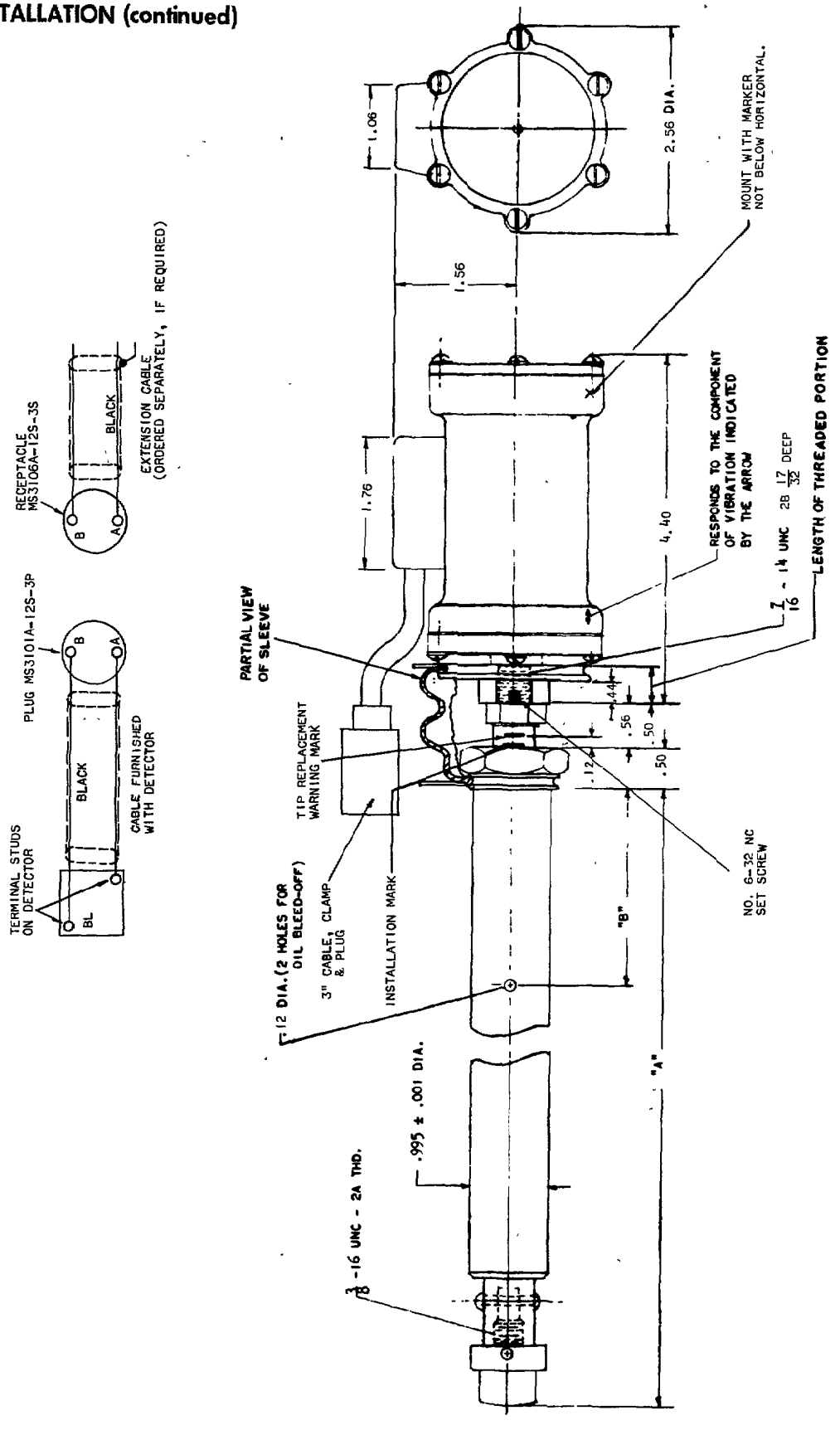


Figure 5 - Outline Dimensions for the 8560K12G702 Vibration Detector With Probe

INSTALLATION (continued)

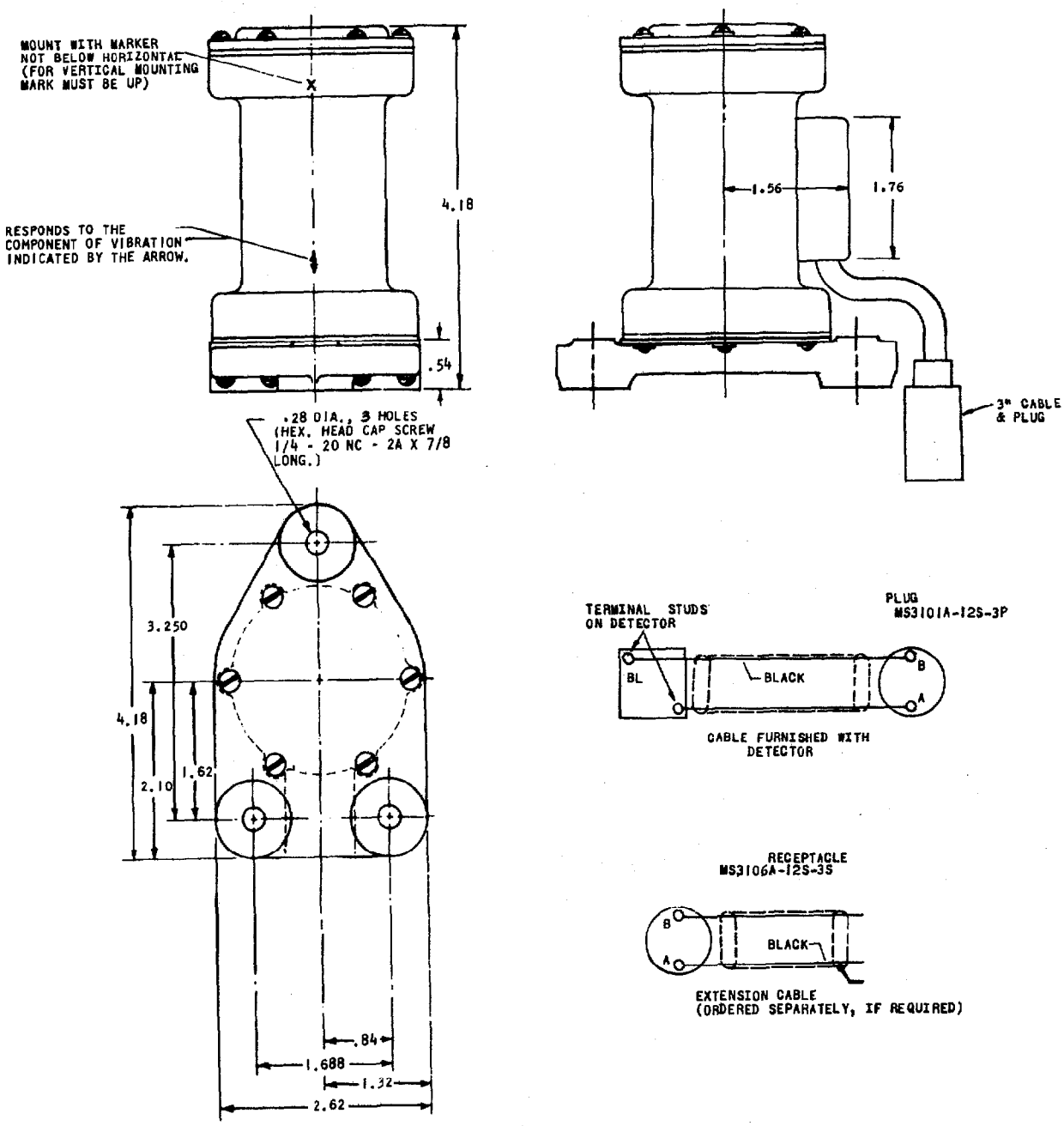


Figure 6 - Outline Dimensions for the 8560K12G703 and 706 Perpendicular Bracket-Mounted Vibration Detectors

INSTALLATION (continued)

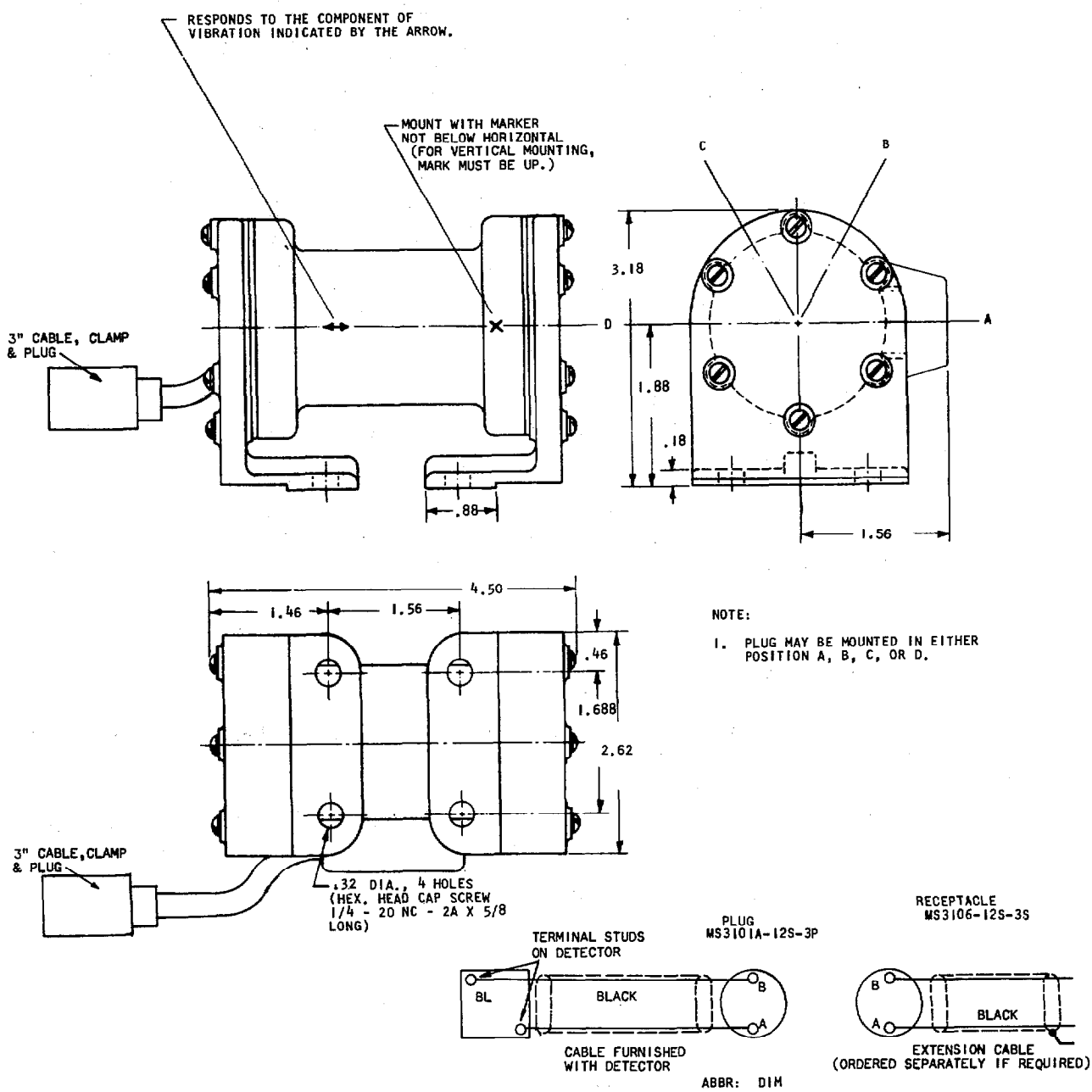


Figure 7 - Outline Dimensions for the 8560K12G704 and 707 Parallel Bracket-Mounted Vibration Detectors

CONNECTIONS

Connections to the electronic cabinet, recording system, or other equipment must be made with shielded cable. At the junction box on the detector case, the black lead must be connected to the terminal marked BLK.

For General Electric turbine supervisory instrument systems, connect as directed in the applicable system instructions. For the GE Cat. No. 4980175 vibration-amplitude recording equipment, connect as directed in instructions GEI-36807 or GEI-73714. For connections to other types of equipment, refer to the applicable instruction book for that equipment, as identified in the introduction section of this publication

For connections between the cabinet or panel-mounted equipment and the cable furnished with the vibration detector, use two-wire shielded cable. Generally a #18 AWG twisted pair covered by a foil shield, with an insulating jacket over the shield is satisfactory. Ground the shield at one point only -- at the electronic cabinet or panel-mounted equipment end. When a long run is made up of two shielded cables joined, the shield of one must be connected to the shield of the other, and the shield for the entire run should be grounded at one end only.

The two-way resistance of the entire cable run, for each vibration detector, should not exceed 35 ohms. The two-way resistance of the cable furnished with the detector will be less than 2 ohms.

When cables are run in strong AC electromagnetic fields or near conductors carrying currents over 50 amperes, the cable should be run in a steel conduit to provide electromagnetic shielding.

MAINTENANCE

SHAFT-VIBRATION DETECTOR

Apply several drops of light machine oil to the outer rider-shaft bearing (at the hex nut of the probe assembly) about once every three months.

The rider-tip assembly must be replaced before the tip replacement warning mark (see Figure 5) becomes flush with the nut on the top of the probe. Also, if the probe is removed for inspection or any other reason, the rider tip must be replaced before reinstallation unless it is known positively that the rider tip is not appreciably worn.

CAUTION

Under no circumstances should the probe be readjusted to show the installation mark once it has disappeared due to wearing of the tip.

To replace the rider-tip assembly, remove the probe from the machine and proceed as follows:

MAINTENANCE (continued)

SHAFT-VIBRATION DETECTOR (continued)

1. File off the peened end of the pin which passes through the rider shaft near the tip assembly and remove the pin.

2. Pass a 1/8-inch drill or a hardened steel rod through the hole in the metal part of the rider-tip assembly, and using this rod and a wrench, unscrew the tip assembly from the bottom of the rider shaft, holding the hex at the opposite end of the rider shaft with a second wrench.

To install a new rider-tip assembly, proceed as follows:

1. Screw the rider-tip assembly into the probe and tighten it as tightly as possible using the 1/8-inch steel rod (or drill) and a wrench.

2. Drill a new pin hole through the shank of the tip holder, drilling through the old holes in the sleeve and rider shaft. Use a 0.1285-inch diameter (#30) drill.

3. Install a new pin and peen the flat end.

For instructions on the lubrication and tip replacement of Dwg. 917C348 probes, manufacturered by the General Electric Large Steam Turbine Department, refer to their instructions GEI-88695.

REPAIR AND ELECTRICAL TROUBLESHOOTING

Check the resistance of the detector leads to ground with an ohmmeter after disconnecting these leads from the associated equipment. The resistance should be several megohms. Measure the resistance between the two pins under the detector cover to check the coil. The resistance should be approximately 190 ohms.

If the detector is mounted on a machine which is running and the approximate amplitude of vibration is known, the detector output may be checked with an electronic voltmeter. If the vibration is sinusoidal and at a frequency corresponding to the speed of the machine, the detector output should be approximately:

$$V(rms) = 0.0000148 \times RPM \times A$$

or

$$V(rms) = 0.000888 \times f \times A$$

where: RPM = speed of machine

A = displacement amplitude of vibration in mils, peak-to-peak (1 mil = 0.001 inch)

f = frequency of vibration in hertz

The wave shape of the detector output, when observed with an oscilloscope, will normally (for most machines) be a symmetrical wave nearly sinusoidal in shape, perhaps with some harmonic content. With shaft vibration detectors, if the wave

MAINTENANCE (continued)

REPAIR AND ELECTRICAL TROUBLESHOOTING (continued)

is markedly non-sinusoidal (peaked or flattened),it is possible that the shaft rider is not following properly. This may be caused by chattering or by the rider shaft being against the stop due to wear of the rider tip. (Refer to INSTALLATION and MAINTENANCE sections on shaft vibration detectors.

Abnormal wave shape may also occur if the suspended coil within the detector bumps against the magnet or pole pieces or is obstructed by dirt or other foreign material within the detector.

REPLACEMENT PARTS

Replacement cables may be ordered by specifying the cable identification shown in Table II.

The rider tip for a probe may be ordered by A5470739G1. Spare pins may be ordered by N312P1212C.

The A5470739G1 replacement tip will not fit the Dwg. No. 917C348 probes. For replacement parts for the 917C348 probes, refer to the General Electric Large Steam Turbine Department, Schenectady, New York.

REPLACING SUPERSEDED TYPES OF DETECTORS

REPLACING CAT. NO. 5470364 DETECTORS

The 8560K12 detectors have the same electrical output and the same mounting arrangement as the Cat. No. 5470364 detectors and can be used as replacement for them. Recommended replacement types are listed in Table IV.

TABLE IV - RECOMMENDED REPLACEMENTS FOR CAT. NO. 5470364 DETECTORS

Old Detector 5470364	Replacement 8560K12	Old Detector 5470364	Replacement 8560K12
G1 thru 11	G702	G46	G702
G12	G703	G47	G702
G13	G704	G48	G702
G14	G702	G49	G706
G15	G702	G50	G707
G16	G702	G51	G705
G17	G703	G52	Not available
G18	G704	G53	G705
G19 thru 31	G702	G54	G702
G32	G703	G55	Not available
G33	G704	G56	G702
G34	G702	G57	G702
G35	G702	G58	G702
G36	G702	G59	G703
G37	G703	G60	G706
G38	G704	G61	G706
G39 thru 43	G702	G62	G705
G44	Not available	G63	G702
G45	Not available		

Replacement cables and/or probes, if required, should be selected from Table II and/or Table III and ordered as separate items

REPLACING SUPERSEDED TYPES OF DETECTORS (continued)

REPLACING OBSOLETE VIBRATION DETECTORS CAT. NO. 9890420 AND 5986021

If an obsolete bracket-mounted vibration detector Cat. No. 9890420G1 thru G5, G7, G9, G10, G11 or Cat. No. 5986021G1 is to be replaced by a 8560K12 detector, follow one of the two alternative installation procedures outlined below:

1. Drill and tap new $\frac{1}{4}$-inch - 20 holes in the surface of the bearing cap, corresponding to the dimensions shown in Figure 6 and 7. Fasten the detector to the machine with the cap screws and lockwashers furnished.

2. If desired, an A5797327G1 adapter plate may be ordered when replacing a Cat. No. 9890420 detector. This adapter plate may be fastened to the existing holes on the machine, and then the detector can be fastened to the adapter plate.

When replacing a Cat. No. 9890420G5 or G11 or Cat. No. 5986021G1 detector with a 8560K12 detector, it will be necessary to connect a 500-ohm $\pm$ 5% wirewound resistor in series with one of the cable leads of the new detector. This resistor should be connected at the panel end of the detector cable.

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